

Lewis Symbols & the Octet Rule



Octet rule — atoms react to reach $8e^-$ in the valence shell (2 for H, He)

Ionic bond — e^- transferred : metal $\rightarrow$ non-metal

Covalent bond — e^- shared : one pair per bond

Lewis Structures of Simple Molecules



Lewis Structure

Step 1 — count e^-

$$N = \sum V_{\text{atoms}} + (-\text{ve charge}) - (+\text{ve charge})$$

- **Step 2** — least EN atom **central** (H, F **never**)
- **Step 3** — single bonds, fill **outer** octets first
- **Step 4** — short of e^- ? make **multiple bonds**

Shortcut — π count

$$\pi = \frac{(\text{octet demand}) - N}{2}$$

$$\text{Octet demand} = 8 \times (\text{atoms}) - 6 \times (\text{H atoms})$$

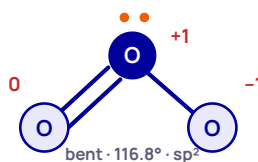
⚠ **Trap**

CO_2 : $N = 4 + 12 = 16$, demand = 24 $\Rightarrow \pi = 4 \Rightarrow$ two $C=O$, not one $C \equiv O$

Formal Charge

$$FC = V - L - \frac{1}{2}B$$

V = valence e^- · L = lone-pair e^- · B = bonding e^-



	O	V	L	B	FC
central		6	2	6	+1
$=O$		6	4	4	0
$-O$		6	6	2	-1

ΣFC must equal the **overall charge** — always check

Octet Rule — 3 Exceptions

Type	Examples	e^-
Incomplete e^- deficient	$BeCl_2$ BF_3 , $AlCl_3$, BH_3	4 6
Expanded needs vacant d	PCl_5 , XeF_2 SF_6 · IF_7	10 12 · 14
Odd-electron paramagnetic	NO · NO_2 ClO_2 · O_2	11 · 17 19

Why period ≥ 3

Period 2 has **no d-orbitals** $\Rightarrow$ no NCI_5 , but PCl_5 ✓

⚠ **High-yield**

$CF_4 + F^- \rightarrow$ **no reaction** but $SiF_4 + 2F^- \rightarrow SiF_6^{2-}$ — d-orbitals

Coordinate Bond

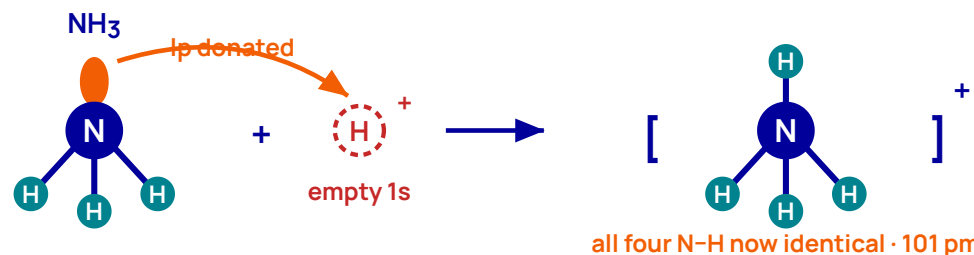
One atom pays for both

- **Donor** brings a **lone pair**
- **Acceptor** brings a **vacant orbital**

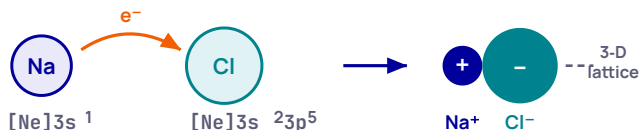
Shown $A \rightarrow B$. Formed differently — then **an ordinary covalent bond**.

⚠ **One-liner**

In NH_4^+ all four N-H bonds are identical — no “dative one”.



Formation & Conditions



Favoured when

Low IE of metal · High -ve ΔH_{eg} of non-metal · High IUI · $\Delta\chi > 1.7$

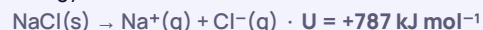
- **Non-directional** & **non-rigid** ⇒ no molecule, no isomerism
- **IUI is the deciding term** – it alone pays back the IE

Lattice Enthalpy

$\Delta_{lattice}H^\circ$

Definition

Energy needed to break 1 mol of an ionic solid into **gaseous ions**.



The only relation you need

$$|U| \propto \frac{z_+ z_-}{r_+ + r_-}$$

Charge beats size: doubling charge quadruples $z_+ z_-$, but radius shifts only ~20%.

Δ Sign convention

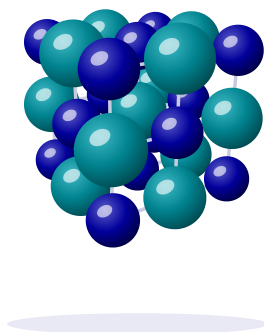
U quoted as **+787** (breaking) or **-787** (forming). Read the arrow, not the number.

Trends in |U|

- **Cation size** ↑ ⇒ IUI ↓
LiF > NaF > KF > RbF > CsF
- **Anion size** ↑ ⇒ IUI ↓
LiF > LiCl > LiBr > LiI
- **Charge beats size**
 $Al_2O_3 > MgO > CaO > NaF > NaCl$

The Rock-Salt Lattice

NaCl · 6:6 coordination



- Na⁺
smaller cation
- Cl⁻
bigger anion

Each ion touches 6 of the other kind

No molecule

the whole crystal is one giant ion pair

Hydration & Solubility

$$\Delta H_{sol} = \Delta H_{hyd} - U$$

Dissolves when $|\Delta H_{hyd}| > |U|$

$$IUI = 787$$

$$|\Delta H_{hyd}| = 770$$

$$\Delta H_{sol}(NaCl) = +17$$

→ endothermic – driven by ΔS

- $\Delta H_{hyd} \propto z/r$ – Li⁺ > Na⁺ > K⁺
- **LiF insoluble** (IUI huge), **LiI soluble**
- **BaSO₄, CaF₂ insoluble** – IUI > $|\Delta H_{hyd}|$

The Ionic Fingerprint

High m.p. / b.p.

Every ion is gripped by all its neighbours – you must beat U to melt it

Hard, yet BRITTLE

Shove one layer and **like charges line up** ⇒ the crystal splits

Conducts only molten / aqueous

Ions are **locked** in the solid; free them and current flows

Dissolves in polar solvents

High ϵ weakens attraction; ΔH_{hyd} pays back U

No isomerism

The bond has **no direction** to hold an arrangement

Solid-State Ionic Forms

autoionisation in the solid

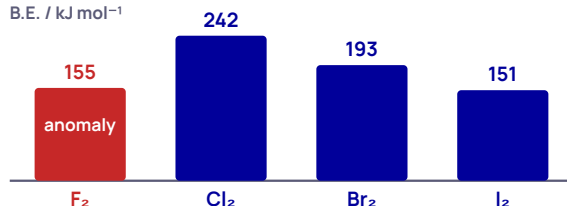
Molecule	Solid exists as	Shapes
PCl_5	$[\text{PCl}_4]^+ [\text{PCl}_6]^-$	$\text{T}_d + \text{O}_h$
PBr_5	$[\text{PBr}_4]^+ \text{Br}^-$	Br^- too big for 6
N_2O_5	$[\text{NO}_2]^+ [\text{NO}_3]^-$	nitronium + nitrate
BrF_3	$[\text{BrF}_2]^+ [\text{BrF}_4]^-$	self-ionises $\Rightarrow$ conducts
XeF_6	$[\text{XeF}_5]^+ \text{F}^-$	F^- -bridged solid

ΔPCl_5 vs PBr_5

PCl_5 gives $[\text{PCl}_6]^-$ but PBr_5 gives plain Br^- . The cation is always the lower-coordinate piece.

Bond Enthalpy

B.E. / kJ mol^{-1}



$\Delta \text{Cl}_2 > \text{F}_2$

F tiny $\Rightarrow$ lone pairs crowd $\Rightarrow$ **lp-lp repulsion** weakens F-F .
Same: $\text{N-N } 163 < \text{P-P } 201 \cdot \text{O-O } 146 < \text{S-S } 266$.

Dissociation $\neq$ mean

CH_4 : 4 unequal steps 427, 439, 452, 347 $\Rightarrow$ mean $1665/4 = 416 \text{ kJ mol}^{-1}$

Bond Order

$$B.O. = \frac{\text{bonds between the 2 atoms}}{\text{resonating structures}}$$

Species	Bonds	B.O.
O_3	$(2+1)/2$	1.5
$\text{CO}_3^{2-}, \text{NO}_3^-$	$4/3$	1.33
SO_3	$6/3$	2.0
C-C benzene	$(2+1)/2$	1.5
RCOO^-	$3/2$	1.5

Bond Length

d / pm

Additivity + electronegativity correction

$$d_{A-B} = r_A + r_B - 0.09 |\chi_A - \chi_B|$$

Schomaker-Stevenson · all lengths in Å. Polar bonds are **shorter** than simple radius addition predicts.

C-C



154 pm

348 kJ

C=C



134 pm

614 kJ

C≡C



120 pm

839 kJ

B.O. $\uparrow \Rightarrow$ length $\downarrow$, energy $\uparrow$ · s-character $\uparrow \Rightarrow$ length $\downarrow$

Learn this row

H-F 92 < H-Cl 127 < H-Br 141 < H-I 161 pm

The 5 Factors

#	Factor	Angle
1	Lone pairs $\uparrow$ on central atom	$\downarrow$
2	EN of central atom $\uparrow$ bp pulled in $\Rightarrow$ crowding $\Rightarrow$ repel	$\uparrow$
3	EN of surrounding atom $\uparrow$ bp pulled away $\Rightarrow$ less crowding	$\downarrow$
4	Multiple bond / bulky group	$\uparrow$
5	Back bonding present	$\uparrow$

Factors 2 & 3 mirror

Where does the bond pair sit? Near the centre $\Rightarrow$ bond pairs collide $\Rightarrow$ angle opens.

The Angle Table

colour = band

BeCl ₂	180°	CO ₂	180°	XeF ₂	180°	I ₃ ⁻	180°	NO ₂ ⁺	180°
BF ₃	120°	SO ₃	120°	NO ₃ ⁻	120°	CO ₃ ²⁻	120°	N(SiH ₃) ₃	120°
(SiH ₃) ₂ O	144°	NO ₂	134°	SO ₂	119.5°	O ₃	117°	NO ₂ ⁻	115°
CH ₄	109.5°	NH ₄ ⁺	109.5°	SO ₄ ²⁻	109.5°	NH ₃	107°	H ₂ O	104.5°
Cl ₂ O	111°	(CH ₃) ₂ O	111°	OF ₂	103°	NF ₃	102°	PCl ₃	100°
PF ₃	97.8°	PH ₃	93.6°	H ₂ S	92°	AsH ₃	91.8°	ClF ₃	87.5°

■ 180° — linear, no lone pair in the way ■ 120–144° — sp² or opened by back bonding ■ 100–119° — sp³ squeezed by lone pairs ■ <100° — heavy central atom keeps near-pure p orbitals

Every Comparison JEE Asks

baseline 85°

Central-atom EN ↓ ⇒ angle ↓

H₂O  104.5°

H₂S  92.1°

H₂Se  91.0°

H₂Te  89.5°

Same story, group 15

NH₃  107.0°

PH₃  93.5°

AsH₃  91.8°

SbH₃  91.3°

Lone pairs ↑ ⇒ angle ↓

CH₄  109.5°

NH₃  107.0°

H₂O  104.5°

Surrounding-atom EN ↑ ⇒ angle ↓

Cl₂O  110.9°

H₂O  104.5°

F₂O  103.2°

△ THE REVERSAL — asked every year

NH₃  107.0°

NF₃  102.0°

PH₃  93.5°

PF₃  97.8° ▲ UP!

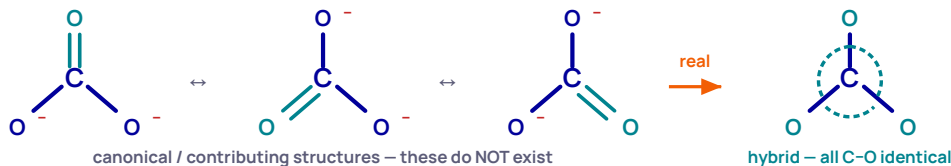
Lone pair > single e⁻ > nothing

NO₂⁺  180.0°

NO₂  134.3°

NO₂⁻  115.4°

Resonance



Conditions for resonance

- **Nuclei must not move** – only e^- shift
- Same number of **unpaired e^-** in every form
- Nearly **equal energy & planar** / conjugated

Resonance energy

$$RE = E_{\text{most stable form}} - E_{\text{hybrid}}$$

Hybrid is **always** more stable $\Rightarrow RE > 0$.

Benzene $\approx 150 \text{ kJ mol}^{-1}$

Most Stable Contributor

- **Max** number of covalent bonds
- **Complete octets** beat extra bonds
- **Min** formal charge / charge separation
- $-ve$ FC on the **more electronegative** atom
- **Like charges never adjacent**

Apply to N_2O



$-ve$ charge sits on O, the more EN atom

Dipole Moment

Definition

$$\mu = q \times d$$

1 D = $3.336 \times 10^{-30} \text{ C}\cdot\text{m}$ = $10^{-18} \text{ esu}\cdot\text{cm}$. Direction: $\delta^+ \rightarrow \delta^-$

Adding two bond moments

$$\mu_R = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos \theta}$$

If the two are equal ($\mu_1 = \mu_2 = \mu$): $\mu_R = 2\mu \cos(\theta/2)$

θ	60°	90°	120°	180°
μ_R	1.73μ	1.41μ	1.00μ	0

$\Delta \mu = 0$ does **NOT** mean non-polar bonds

CO_2 : **two polar C=O**, exactly opposed (180°) $\Rightarrow$ **molecule** non-polar, **bonds** are not.

Dipole Exception

$\text{CH}_3\text{Cl} > \text{CH}_3\text{F}$

F is more EN $\Rightarrow$ you expect CH_3F larger. It isn't:

CH_3Cl 1.87 D > CH_3F 1.85 D

The reason

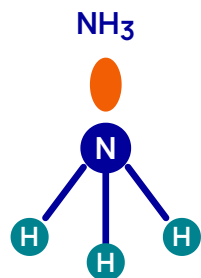
$$\mu = q \times d$$

C-Cl is much longer than C-F $\Rightarrow$ larger **d** offsets the smaller **q** $\Rightarrow \text{CH}_3\text{Cl}$ wins.

Δ **Not electronegativity alone**

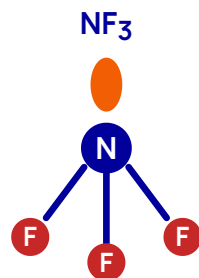
Dipole moment is charge $\times$ **distance**. Bond length is doing the work here.

NH_3 vs NF_3



bond $\uparrow$ lp $\uparrow$
SAME way
 $\Rightarrow$ ADD

$\mu = 1.47 \text{ D}$ — LARGE



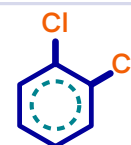
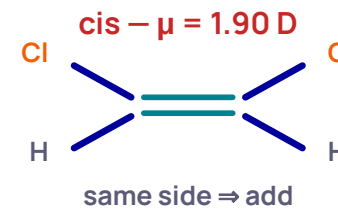
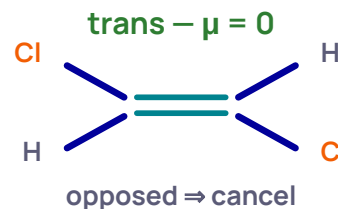
bond $\downarrow$ lp $\uparrow$
OPPOSE
 $\Rightarrow$ CANCEL

$\mu = 0.23 \text{ D}$ — TINY

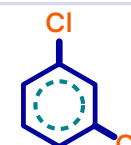
Does the lp moment help the bond moments, or fight them?

Geometry Decides

cis/trans · o/m/p



ortho $\theta = 60^\circ$ · $\mu = 2.50 \text{ D}$



meta $\theta = 120^\circ$ · $\mu = 1.72 \text{ D}$



para $\theta = 180^\circ$ · $\mu = 0$

$$\mu_R = 2\mu \cos(\theta/2), \mu(\text{C-Cl}) \approx 1.5 \text{ D} \Rightarrow \text{ortho } 2.60 \cdot \text{meta } 1.50 \cdot \text{para } 0$$

% Ionic Character from μ

Real vs 100% ionic

$$\% \text{ ionic} = \frac{\mu_{\text{obs}}}{\mu_{\text{ionic}}} \times 100$$

Shortcut: a full electronic charge across $1 \text{ \AA}$ gives $4.8 \text{ D} \Rightarrow \mu_{\text{ionic}}(\text{D}) = 4.8 \times d(\text{\AA})$

Worked — HCl

$d(\text{H-Cl})$	1.27 $\text{\AA}$	given
μ_{ionic}	6.10 D	4.8×1.27
μ_{obs}	1.03 D	measured
% ionic	16.9 %	$1.03/6.10$

Hannay-Smyth

% ionic character

From electronegativity

$$\% \text{ ionic} = 16 \Delta\chi + 3.5 \Delta\chi^2$$

$$\Delta\chi = |\chi_A - \chi_B|$$

From dipole moment

$$\% \text{ ionic} = \frac{\mu_{\text{obs}}}{\mu_{\text{ionic}}} \times 100$$

⚠ Two different routes

One uses ΔEN , the other observed vs 100%-ionic μ . Don't mix the formulas.

The $\mu = 0$ Club

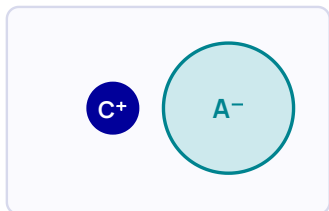
Identical atoms + no lp on the centre (or symmetric lp) $\Rightarrow$ every bond moment cancels.

CO_2 CS_2 BeCl_2 BeF_2 BF_3 BCl_3 CH_4 CCl_4 SiF_4 PCl_5 SF_6 XeF_2 XeF_4 C_2H_4 C_2H_2 C_6H_6 $\text{trans-C}_2\text{H}_2\text{Cl}_2$ $\text{p-C}_6\text{H}_4\text{Cl}_2$

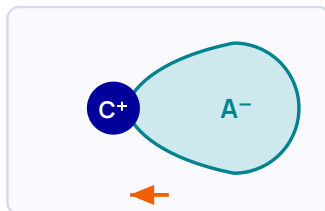
$\mu \neq 0$ — a lp, a bend or an odd substituent breaks it (D):

$\text{HF } 1.78$ $\text{HCl } 1.07$ $\text{HBr } 0.79$ $\text{HI } 0.38$ $\text{H}_2\text{O } 1.85$ $\text{H}_2\text{S } 0.95$ NH_3 1.47 NF_3 0.23 SO_2 1.63 $\text{CH}_3\text{Cl } 1.86$ CH_2Cl_2 1.60 CHCl_3 1.04 $\text{cis-C}_2\text{H}_2\text{Cl}_2$ 1.90 $\text{o-C}_6\text{H}_4\text{Cl}_2$ 2.50

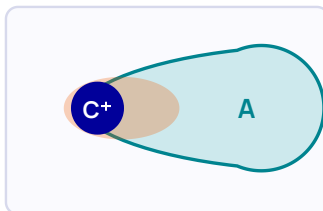
Polarisation — The Picture



Symmetrical ⇒ IONIC



Cloud pulled ⇒ POLARISED e⁻ shared ⇒ COVALENT



Cation — polarising

$$\text{P.P.} \propto \frac{z_+}{r_+^2} \quad \phi = \frac{z_+}{r_+}$$

ϕ = Cartledge ionic potential

Anion — polarisable

∝ size & ∝ -ve charge

Loosely held e⁻ cloud distorts easily

Fajans' 4 Rules

#	Condition	Example
1	Small cation	$\text{Li}^+ > \text{Na}^+ > \text{K}^+$
2	Large anion	$\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$
3	High charge on either	$\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$
4	Pseudo noble-gas cation	$\text{Cu}^+ > \text{Na}^+$

Rule 4 — $18e^- > 8e^-$

Pseudo ($ns^2np^6nd^{10}$) → d-electrons screen poorly ⇒ higher Z_{eff} ⇒ stronger pull

△ Classic pair

$r(\text{Cu}^+) 96 \approx r(\text{Na}^+) 95 \text{ pm}$, same charge — yet **CuCl covalent**, **NaCl ionic**

Consequences

Covalent character ↑ ⇒

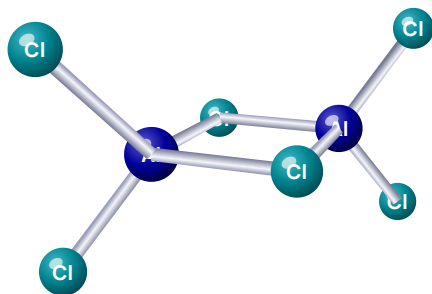
m.p. / b.p.	↓
Solubility in H_2O	↓
Solubility in C_6H_6 , ether	↑
Thermal stability	↓
Colour intensity	↑
Conductivity (molten)	↓

△ Trap

Explains covalent character in an IONIC bond — never the reverse.

Al_2Cl_6

the dimer Fajans predicts



two tetrahedral Al sharing a Cl-Cl edge

Each bridging Cl gives a lone pair to the other Al ⇒ both Al reach an octet.

Orders You Must Know

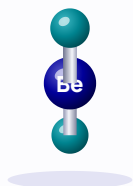
Series — covalent character decreases →	Rule	Evidence
$\text{LiCl} > \text{NaCl} > \text{KCl} > \text{RbCl} > \text{CsCl}$	1	LiCl dissolves in ethanol
$\text{LiI} > \text{LiBr} > \text{LiCl} > \text{LiF}$	2	LiF ionic & insoluble
$\text{AlCl}_3 > \text{MgCl}_2 > \text{NaCl}$	3	AlCl_3 sublimes at 192°C
$\text{FeCl}_3 > \text{FeCl}_2 \quad \cdot \quad \text{SnCl}_4 > \text{SnCl}_2$	3	higher halide is volatile
$\text{AgI} > \text{AgBr} > \text{AgCl} > \text{AgF}$	2,4	AgF soluble; AgI yellow
$\text{CuCl} > \text{NaCl} \quad \cdot \quad \text{ZnCl}_2 > \text{CaCl}_2$	4	$18e^-$ core

Colour deepens with polarisation: **AgCl** white → **AgBr** pale yellow → **AgI** yellow

The Shape Gallery

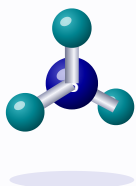
navy = centre · teal = bonded · orange = lone pair

S.N. 2



Linear
 BeCl_2
 180°

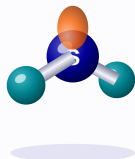
S.N. 3



Trigonal planar
 BF_3
 120°

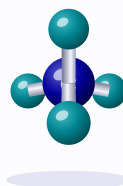
S.N. 3

1 lp



Bent
 SO_2
 119°

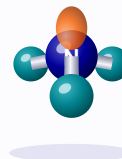
S.N. 4



Tetrahedral
 CH_4
 109.5°

S.N. 4

1 lp



Pyramidal
 NH_3
 107°

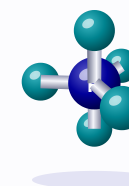
S.N. 4

2 lp



Bent
 H_2O
 104.5°

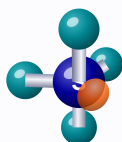
S.N. 5



Trig. bipyram.
 PCl_5
 $120^\circ/90^\circ$

S.N. 5

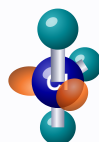
1 lp



See-saw
 SF_4
 $173^\circ/102^\circ$

S.N. 5

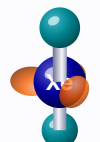
2 lp



T-shaped
 ClF_3
 87.5°

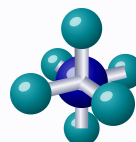
S.N. 5

3 lp



Linear
 XeF_2
 180°

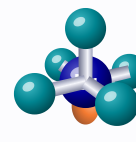
S.N. 6



Octahedral
 SF_6
 90°

S.N. 6

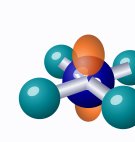
1 lp



Sq. pyramidal
 BrF_5
 84.8°

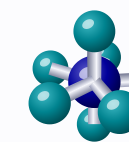
S.N. 6

2 lp



Square planar
 XeF_4
 90°

S.N. 7

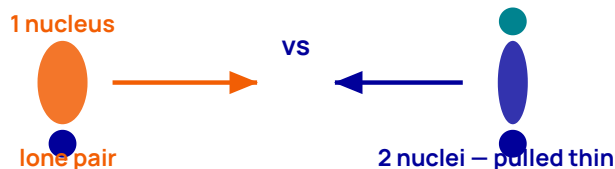


Pent. bipyram.
 IF_7
 $72^\circ/90^\circ$

Repulsion — The One Rule

$$\text{lp-lp} > \text{lp-bp} > \text{bp-bp}$$

Held by **one** nucleus $\Rightarrow$ fatter, closer cloud $\Rightarrow$ shoves harder.



- Each lp shrinks the bond angle by roughly **2–2.5°**
- Multiple bonds** = **one** domain, but repel **more** than a single

Geometry $\neq$ Shape

Species	Geometry (e ⁻ pairs)	Shape (atoms)
CH ₄	Tetrahedral	Tetrahedral
NH ₃	Tetrahedral	Pyramidal
H ₂ O	Tetrahedral	Bent
SF ₄	Trig. bipyramidal	See-saw
XeF ₂	Trig. bipyramidal	Linear
XeF ₄	Octahedral	Square planar

Geometry counts lp. **Shape** ignores them — you only see atoms.

Where Does the Lone Pair Sit?

In sp³d, count the **90° lp-bp** contacts — fewer wins.

lp position	90° lp-bp	Verdict
Equatorial	2	chosen
Axial	3	rejected

Consequences

SF₄, ClF₃, XeF₂: lp **always equatorial** $\Rightarrow$ axial bonds bend $\Rightarrow$ 173°, 87.5°, 180°.

△ In sp³d² (octahedral)

6 sites equivalent $\Rightarrow$ 1st lp anywhere, 2nd goes **trans** $\Rightarrow$ XeF₄ square planar, never see-saw.

Drago's Rule

the ~90° / pure-p rule

Central atom of **period ≥ 3** bonded to **low-EN** atoms (H) $\Rightarrow$ **no hybridisation** $\Rightarrow$ bonds through **pure p** $\Rightarrow$ angle $\rightarrow$ **90°**.

Hydride	Angle	Hydride	Angle
NH ₃	107°	H ₂ O	104.5°
PH ₃	93.5°	H ₂ S	92°
AsH ₃	91.8°	H ₂ Se	91°
SbH ₃	91.3°	H ₂ Te	90°

△ Don't force sp³

The near-90° angle is the **signal** hybridisation was skipped. NH₃/H₂O still hybridise (small, EN centre).

Odd-Electron Species

NO₂, ClO₂

SN = $\frac{1}{2}[V + M - c + a]$ assumes pairing. An **unpaired e⁻** repels **less than a lone pair** — about "half a lp."

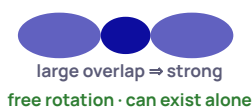
Species	On centre	Angle
NO ₂ ⁺	no lp	180°
NO ₂	1 odd e ⁻	134°
NO ₂ ⁻	full lp	115°
ClO ₂	1 odd e ⁻	117.5°

△ Half a lone pair

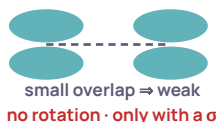
Angle lands **between** the no-lp and full-lp cases: **180° > 134° > 115°**.

Orbital Overlap — σ vs π

σ — head-on / axial



π — sidewise / lateral



Overlap $\Rightarrow$ strength

$s-s < s-p < p-p$ (σ) · and always $\sigma > \pi$

△ Counting bonds

Single 1σ · Double $1\sigma+1\pi$ · Triple $1\sigma+2\pi$ · **first bond always σ**

VBT — What It Cannot Do

Postulates (Heitler–London, Pauling–Slater)

- **Half-filled AOs, opposite spins**
- **Directional** · strength $\propto$ overlap
- Energy **minimum** at the bond length

△ Four failures

- **O_2 paramagnetic** — VBT says all paired
- Cannot rank $O_2^+ / O_2 / O_2^- / O_2^{2-}$, no fractional B.O.
- Cannot explain **He_2^+ ✓, He_2 ✗**
- Bonds **localised** $\Rightarrow$ resonance is a patch, not a prediction

Steric Number

$$H = \frac{1}{2} [V + M - C + A]$$

V valence e^- of centre · M monovalent atoms (H, F, Cl, Br, I) · C +ve · A -ve charge

△ Easy to miss

Terminal **O and S contribute ZERO** to M — they are divalent.

Species	$\frac{1}{2}[V + M - C + A]$	H	Hybrid
NH_4^+	$\frac{1}{2}[5+4-1]$	4	sp^3
SO_4^{2-}	$\frac{1}{2}[6+0+2]$	4	sp^3
I_3^-	$\frac{1}{2}[7+2+1]$	5	sp^3d
XeF_4	$\frac{1}{2}[8+4]$	6	sp^3d^2
NO_2^+	$\frac{1}{2}[5+0-1]$	2	sp
XeF_6	$\frac{1}{2}[8+6]$	7	sp^3d^3

Hybridisation → Geometry

H	Hybrid	d-orbitals used	Examples
2	sp	—	$BeCl_2$, CO_2 , C_2H_2
3	sp^2	—	BF_3 , SO_3 , NO_3^-
4	sp^3	—	CH_4 , NH_4^+ , SO_4^{2-}
5	sp^3d	d_{z^2}	PCl_5 , SF_4 , XeF_2
6	sp^3d^2	d_{z^2} , $d_{x^2-y^2}$	SF_6 , XeF_4 , BrF_5
7	sp^3d^3	d_{z^2} , $d_{x^2-y^2}$, d_{xy}	IF_7 , XeF_6

Rules · inner-orbital case

Comparable energy · **orbitals** hybridise, **not electrons** · $dsp^2 \rightarrow$ sq. planar $[Ni(CN)_4]^{2-}$

s-character → Angle

Hybrid	% s	θ
sp	50	180°
sp^2	33.3	120°
sp^3	25	$109^\circ 28'$

% s $\uparrow \Rightarrow$ angle $\uparrow$

Effects of ↑ s-character

Electronegativity of C

$sp \ 3.2 > sp^2 \ 2.75 > sp^3 \ 2.5$

Acidity of C-H (pK_a)

$HC\equiv CH \ 25 > H_2C=CH_2 \ 44 > C_2H_6 \ 50$

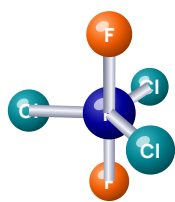
C-H bond length / pm

$sp \ 106 < sp^2 \ 108 < sp^3 \ 110$

- Bond **strength** ↑, **length** ↓
- Basicity** ↓ — lp held tighter · pyridine (sp^2) < piperidine (sp^3)

Bent's Rule

sp^3d · axial vs equatorial



PCl_3F_2

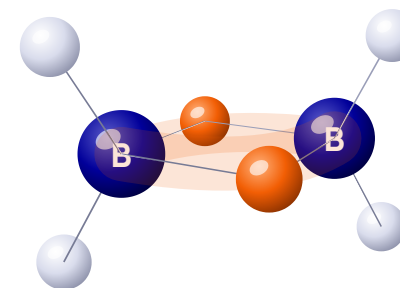
- s-character** → electropositive; **p-character** → electronegative atoms
- Equatorial** = sp^2 (more s) ⇒ shorter, stronger
- Axial** = p-rich ⇒ longer, weaker
- Electronegative** → **axial** · lone pairs → equatorial
- PCl_3F_2 : 2 F axial · SF_4 axial 1.65 > eq 1.55 Å

△ Two different rules

VSEPR puts **lone pairs** equatorial (room). Bent puts **electronegative bonded atoms** axial. Don't merge them.

Bridge Bonding

B_2H_6 · 3c-2e banana bonds



Only **12 valence e^-** for 8 B-H contacts ⇒ electron-deficient. B is sp^3 ; **no direct B-B bond**.
4 terminal B-H = $2c-2e$ ($8e^-$) · **2 bridging** B-H-B = $3c-2e$ ($4e^-$) ⇒ 12 ✓

△ No B-B bond

The borons are held together **only** through the H bridges — never draw B=B or 4 normal B-H.

LCAO

All three must hold

Comparable energy · proper symmetry about the axis · appreciable overlap

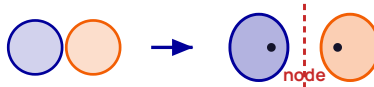
n AOs in $\Rightarrow n$ MOs out

SAME phase $\rightarrow$ ADD $\rightarrow \sigma$

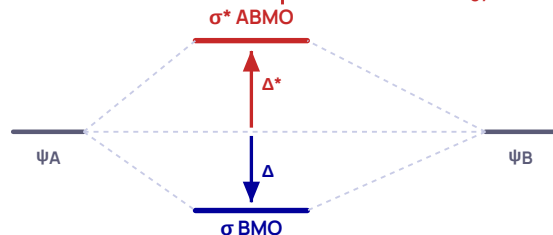


no nodal plane
density $\uparrow$ between nuclei
energy LOWERED

OPPOSITE phase $\rightarrow$ SUBTRACT $\rightarrow \sigma^*$



nodal plane $\perp$ axis
density pushed OUT
energy RAISED

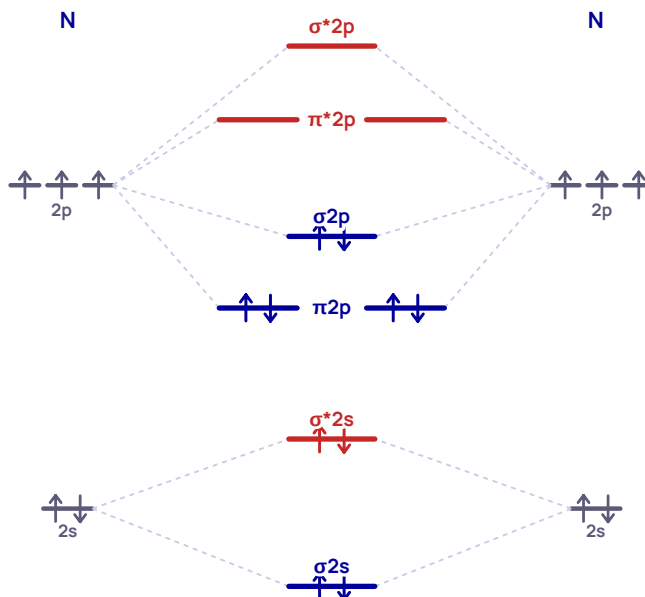


Δ The asymmetry

$\Delta^* > \Delta$ — ABMO rises more than BMO falls $\Rightarrow$ **He₂ X** · He₂⁺ ✓ (B.O. 0.5).

Ladder ①: total e⁻ ≤ 14

Used by: Li₂ · Be₂ · B₂ · C₂ · N₂ | CO, CN⁻, NO⁺

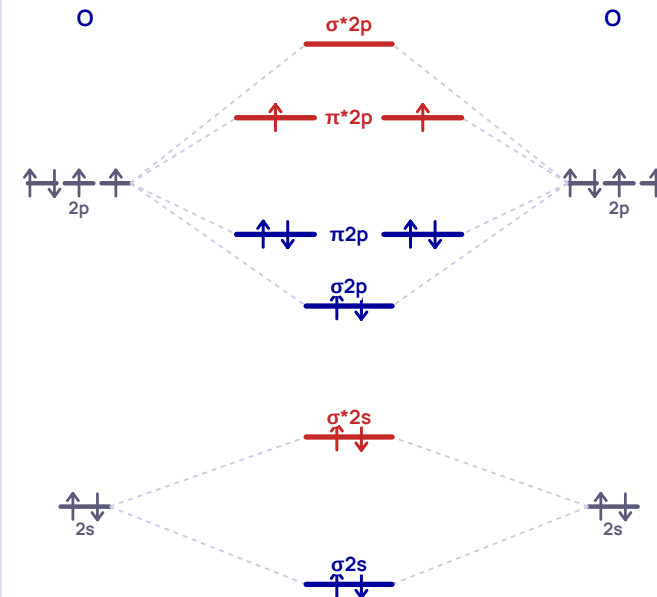


KK $\sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 \pi 2p_y^2 \sigma 2p_z^2$

$N_b = 8 \cdot N_a = 2 \Rightarrow$ **B.O. = $(8-2)/2 = 3$ · 0 unpaired $\Rightarrow$ DIAMAGNETIC**

Ladder ②: total e⁻ > 14

Used by: O₂ · F₂ · Ne₂ | NO, O₂⁻, O₂²⁻



KK $\sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^1 \pi^* 2p_y^1$

$N_b = 8 \cdot N_a = 4 \Rightarrow$ **B.O. = $(8-4)/2 = 2$ · 2 unpaired $\Rightarrow$ PARAMAGNETIC**

Bond Order

Count, subtract, halve

$$\text{B.O.} = \frac{N_b - N_a}{2}$$

N_b = e⁻ in bonding MOs · N_a = e⁻ in antibonding MOs

If B.O.	Then	Why
$\uparrow$	Energy $\uparrow$ · length $\downarrow$ · stability $\uparrow$	more net glue
$= 0$	Species does NOT exist	He ₂ , Be ₂ , Ne ₂

Fractional B.O. is legal: H₂⁺ 0.5 · NO 2.5 · O₂⁻ 1.5

Δ K electrons?

No. $\sigma 1s^2 + 2 N_b$ and $\sigma^* 1s^2 + 2 N_a$ cancel $\Rightarrow$ KK or full count, same B.O.

HOMO / LUMO

reading the ladder

- HOMO** = highest **filled** MO (top occupied level)
- LUMO** = lowest **empty** MO (first vacant above)

	HOMO	LUMO
N ₂	$\sigma 2p$	$\pi^* 2p$
O ₂	$\pi^* 2p$	$\sigma^* 2p$

Ionisation strips the HOMO e⁻ · reduction fills the LUMO.

Δ N₂ HOMO is $\sigma 2p$

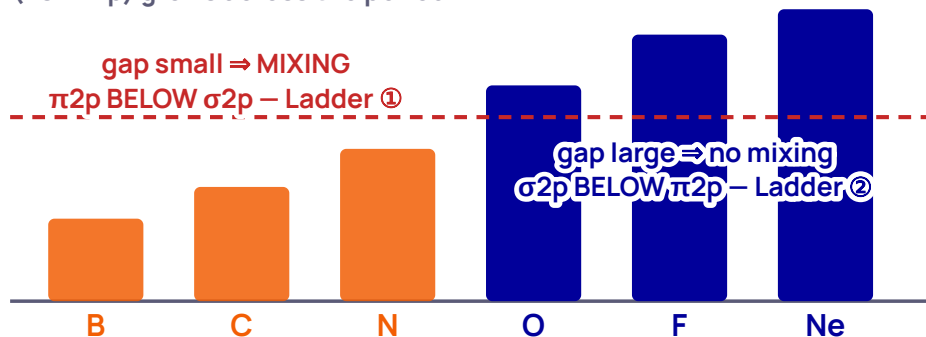
s-p mixing lifts $\sigma 2p$ above $\pi 2p$ (Ladder ①). For O₂/F₂ the order reverts.

Why the σ - π Crossover?

2s & 2p_z: **same σ symmetry** $\Rightarrow$ mix $\Rightarrow$ **$\sigma 2p_z$ UP**, $\sigma 2s$ down.

$\pi 2p$ – different symmetry $\Rightarrow$ untouched.

ΔE (2s $\rightarrow$ 2p) grows across the period $\rightarrow$



Cause: Z_{eff} $\uparrow$ across the period $\Rightarrow$ 2s (better penetration) is pulled down **faster** than 2p $\Rightarrow$ the gap widens.

Every Period-1/2 Diatomic

Species	e ⁻	Ldr	Valence configuration	$N_b - N_a$	B.O.	n	Magnetic	Real?
H ₂ ⁺	1	1s	$\sigma 1s^1$	1-0	0.5	1	PARA	✓
H ₂	2	1s	$\sigma 1s^2$	2-0	1	0	DIA	✓
He ₂ ⁺	3	1s	$\sigma 1s^2 \sigma^* 1s^1$	2-1	0.5	1	PARA	✓
He ₂	4	1s	$\sigma 1s^2 \sigma^* 1s^2$	2-2	0	0	–	✗
Li ₂	6	①	KK $\sigma 2s^2$	4-2	1	0	DIA	✓
Be ₂	8	①	KK $\sigma 2s^2 \sigma^* 2s^2$	4-4	0	0	–	✗
B ₂	10	①	KK $\sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^1 \pi 2p_y^1$	6-4	1	2	PARA	✓
C ₂	12	①	KK $\sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 \pi 2p_y^2$	8-4	2	0	DIA	✓
N ₂	14	①	KK $\sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 \pi 2p_y^2 \sigma 2p_z^2$	10-4	3	0	DIA	✓
O ₂	16	②	KK $\sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^1 \pi^* 2p_y^1$	10-6	2	2	PARA	✓
F ₂	18	②	KK $\sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^2 \pi^* 2p_y^2$	10-8	1	0	DIA	✓
Ne ₂	20	②	KK $\sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^2 \pi^* 2p_y^2 \sigma^* 2p_z^2$	10-10	0	0	–	✗

Magnetism

Spin-only moment

$$\mu = \sqrt{n(n+2)} \text{ BM} \quad (n = \text{unpaired } e^-)$$

n	0	1	2	3
μ / BM	0	1.73	2.83	3.87
type	DIA	PARA	PARA	PARA

✓ The O₂ test case

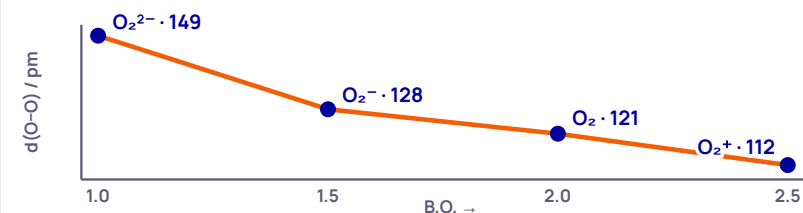
VBT: O=O, all paired $\Rightarrow$ diamagnetic. **Wrong.**

MOT: $\pi^* 2p_x^1 \pi^* 2p_y^1 \Rightarrow$ 2 unpaired $\Rightarrow$ paramagnetic – liquid O₂ sticks to a magnet.

B₂ para, C₂ dia – only Ladder ① gets both right; ② reverses them.

The O₂ⁿ Family

Species	e ⁻	$\pi^* 2p$ filling	B.O.	d / pm	Magnetic
O ₂ ⁺ dioxygenyl	15	$\uparrow -$	2.5	112	PARA
O ₂	16	$\uparrow \uparrow$	2.0	121	PARA
O ₂ ⁻ superoxide	17	$\uparrow \downarrow \uparrow$	1.5	128	PARA
O ₂ ²⁻ peroxide	18	$\uparrow \downarrow \uparrow \downarrow$	1.0	149	DIA

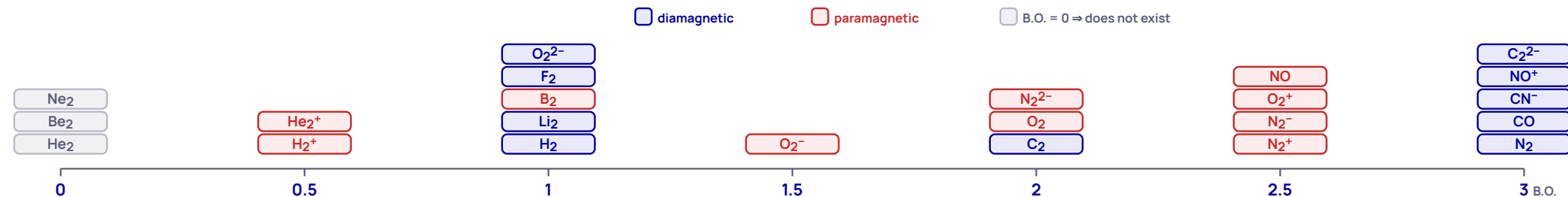


Added e⁻ $\rightarrow$ $\pi^* \Rightarrow$ B.O. $\downarrow$

B.O. / stability / energy: O₂⁺ > O₂ > O₂⁻ > O₂²⁻

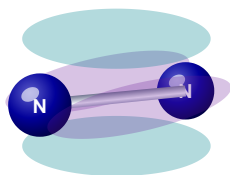
Length: O₂⁺ < O₂ < O₂⁻ < O₂²⁻

The Bond-Order Ladder



What a Triple Bond Looks Like

$$N \equiv N \cdot 1\sigma + 2\pi$$

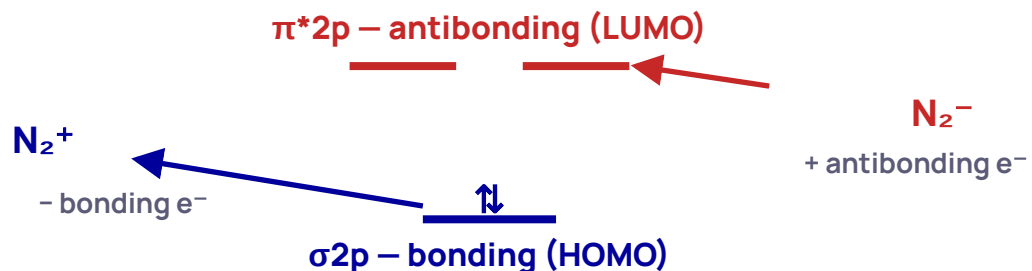


- σ bond**
head-on overlap along the axis – the strong one
- π bonds $\times 2$**
sideways overlap – two of them, at 90° to each other

The first bond is always σ
every extra bond is π

MOT Tie-breaker

$$N_2^+ > N_2^- \cdot \text{both B.O. } 2.5$$



Both drop from B.O. 3 $\rightarrow$ 2.5, but **where** the e^- moved differs. Antibonding e^- hurts more.
 N_2^+ lose bonding $\sigma 2p \Rightarrow$ B.E. $\uparrow$, bond $\downarrow$ · N_2^- gain antibonding $\pi^* 2p$

Δ Equal B.O. $\neq$ equal stability

Ask which MO the electron entered or left – bonding or antibonding.

Sulphur Vapour

S_2 is paramagnetic

Above $\sim 1000^\circ\text{C}$, S vapour is mainly S_2 – **paramagnetic**, just like O_2 .
 2 unpaired e^- in π^* · $S=S$ double bond (B.O. 2)

Same valence MO pattern as O_2 .

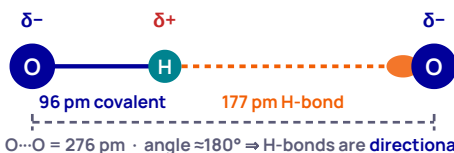
Δ Only the vapour

Solid sulphur is S_8 (puckered ring, diamagnetic). Only hot S_2 vapour is paramagnetic.

What Counts as an H-Bond?

Both, or no H-bond

H on a **small, highly EN** atom – **F, O, N only** · acceptor Y offers a **lone pair**

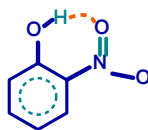


Interaction	kJ mol^{-1}
van der Waals	0.1 – 10
Hydrogen bond	5 – 40
O-H covalent bond	463



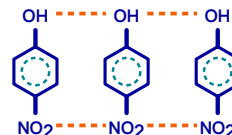
Intra vs Inter

INTRA – chelation



o-nitrophenol · locked in a 6-ring

INTER – chains



p-nitrophenol · every molecule links up

Property	INTRA (ortho)	INTER (para)
m.p. / b.p.	LOW	HIGH
Volatility	HIGH	LOW
Solubility in H_2O	LOW	HIGH
Steam distillable	✓ yes	✗ no

The Ortho Effect

Compound	H-bond	m.p. / $^\circ\text{C}$
o-nitrophenol	INTRA	45
p-nitrophenol	INTER	114
salicylic acid	INTRA	159
p-hydroxybenzoic acid	INTER	215

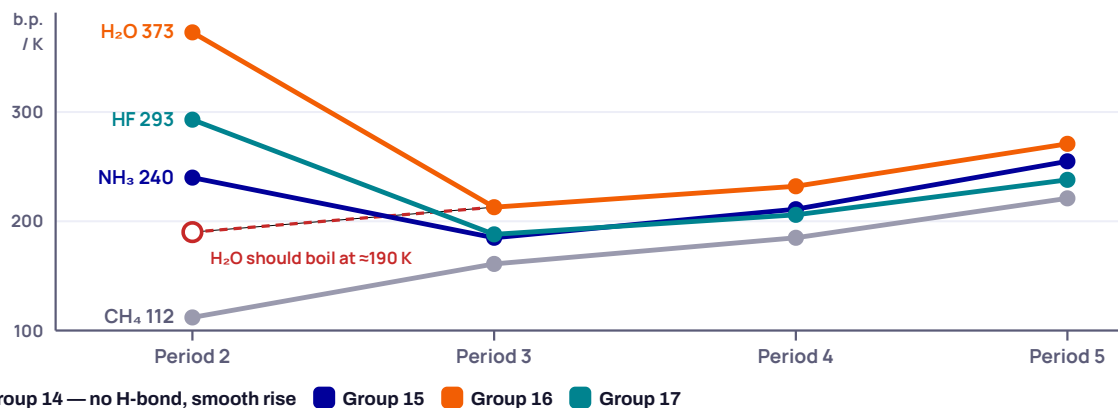
Why ortho

Ring must close at **5–6 members**, unstrained. **meta/para**: too far $\Rightarrow$ bond **outward**.

⚠ Not always ortho

Intra H-bond stabilises the conjugate base $\Rightarrow$ **salicylic acid more acidic** than its p-isomer.

The Boiling-Point Anomaly

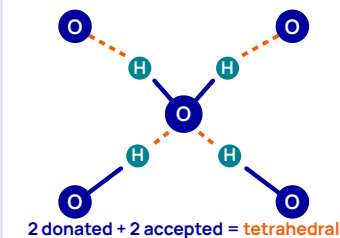


Why the Jump?

	Expected	Actual	Jump
H_2O	$\approx 190 \text{ K}$	373 K	+183
HF	$\approx 150 \text{ K}$	293 K	+143
NH_3	$\approx 150 \text{ K}$	240 K	+90

$\text{H}_2\text{O} > \text{HF}$ despite $\text{EN}(\text{F}) \uparrow - 2\text{H} + 2\text{lp} \Rightarrow 4 \text{ H-bonds} \Rightarrow 3\text{-D net}$. HF: $1\text{H} \Rightarrow 2 \Rightarrow$ chains.

Ice vs water



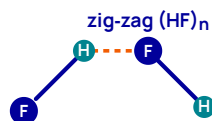
ICE — open cage, $\approx 34\%$ empty
 $\rho = 0.92 \text{ g cm}^{-3} \Rightarrow$ it floats

▼ melt

WATER — cage collapses, packs tight
 $\rho = 1.00 \text{ g cm}^{-3}$ · max at 4°C

Aquatic life survives winter on this one fact

HF — the Strongest H-Bond



bifluoride $[\text{F}\cdots\text{H}\cdots\text{F}]^-$



symmetric — H sits exactly in the middle
 $\approx 165 \text{ kJ mol}^{-1}$ — the strongest H-bond known

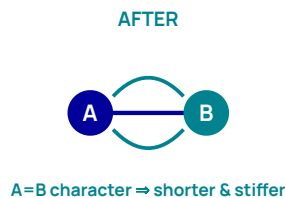
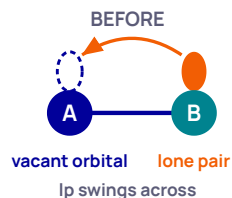
⚠ Why HF is a weak acid

Strong H-F + ion-pairing locks H^+ away $\Rightarrow \text{HF} < \text{HCl} < \text{HBr} < \text{HI}$ in acidity.

Back Bonding

Both partners

- One atom has a **lone pair** to give
 - Its neighbour has a **vacant orbital** to take it
- Result: a **partial π bond** on top of the σ bond.

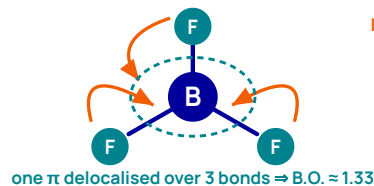


Type	Donor $\rightarrow$ Acceptor	Example
$p\pi-p\pi$	2p 1p $\rightarrow$ vacant 2p	BF_3 , B(OR)_3
$p\pi-d\pi$	2p 1p $\rightarrow$ vacant 3d	$\text{N(SiH}_3)_3$, POCl_3

BF_3

B: empty 2p

F: 3 lone pairs

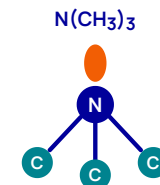


B-F bond	pm
Calculated single bond $r_B + r_F$	152
Observed in BF_3	130

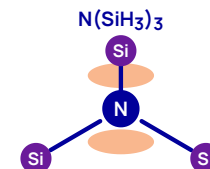
Δ Lewis acidity runs **BACKWARDS**

$\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$ — EN predicts the reverse.
F 2p $\approx$ B 2p $\Rightarrow$ best overlap $\Rightarrow$ strongest back bonding $\Rightarrow$ B least needy.

$\text{N(SiH}_3)_3$ vs $\text{N(CH}_3)_3$



lp ACTIVE $\Rightarrow$ sp^3 pyramidal $\cdot 108^\circ$



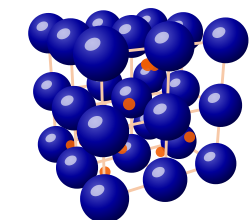
lp DONATED to Si 3d $\Rightarrow$ sp^2 PLANAR $\cdot 120^\circ$

Pair	Angle	Shape
$\text{N(CH}_3)_3$	108°	pyramidal
$\text{N(SiH}_3)_3$	120°	PLANAR
$(\text{CH}_3)_2\text{O}$	111°	bent
$(\text{SiH}_3)_2\text{O}$	144°	opened up

lp donated away $\Rightarrow$ stops repelling $\Rightarrow$ angle opens, basicity collapses: $\text{N(SiH}_3)_3 \ll \text{N(CH}_3)_3$.

Metallic Bonding

the electron sea, in 3D



- Kernel**
fixed in the lattice
- Valence e^-**
free to roam

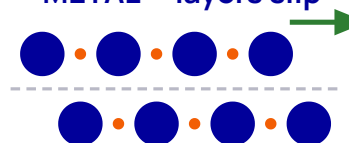
Strength $\propto$
valence $e^- \uparrow \cdot$ kernel $\downarrow$

Na 371 < Mg 923 < Al 933 K

Li > Na > K > Rb > Cs

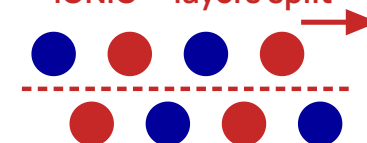
Why Metals Behave That Way

METAL — layers slip



sea re-forms $\Rightarrow$ **BENDS**
malleable & ductile

IONIC — layers split



like charges meet $\Rightarrow$ **CRACKS**
hard but brittle

Property	Because the e^- are...
Conducts electricity	free to drift in a field
Conducts heat	mobile carriers of energy
Lustrous	able to absorb & re-emit light